Speaker Notes

Dynamic Urban Network Topologies and Epidemic Dynamics

A Comparative Study of Static vs Adaptive Contact Networks for COVID-19 Modeling

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Companion notes for the 17-slide deck. Each section gives the talking points for the corresponding slide, the critical numbers to know, and the most likely audience questions with prepared answers.

Slide 1 - Cover - Research Topic Introduction

What to Say

- Good [morning/afternoon]. I'm Yasir Abbas, Group J4133, ITMO University.
- My scientific supervisor is Prof. Vasiliy Leonenko.
- Today I'm presenting my work on Dynamic Urban Network Topologies and Epidemic Dynamics - a comparative study of static vs adaptive contact networks for COVID-19 modeling.
- The work bridges network theory, urban science, and epidemiology.

Key Numbers / Facts

! *Programme: Applied Mathematics and Computer Science 01.04.02*

Possible Questions and Answers

Q: Why did you choose this topic?

A: COVID-19 demonstrated that classical epidemic models miss a critical feedback loop - people change behavior when they perceive risk. No prior study systematically compared static vs dynamic predictions across multiple urban topologies with real-data validation, so the work fills a clear gap.

Q: How does this build on first-semester work?

A: Semester 1 proved different urban topologies produce different outbreak patterns under static assumptions. This semester extends that by adding dynamic behavior and validating against real St. Petersburg data.

Slide 2 - Statement of the Problem

What to Say

- Traditional SEIR-type models assume the contact network is static - that human behavior is constant throughout an outbreak. This is mathematically convenient but empirically wrong.
- In reality there is a two-way coupling: infections change behavior, and changed behavior changes future infections.
- My scientific question is whether incorporating dynamic contact patterns - weekly rhythms plus threshold-based behavioral adaptation - materially changes predicted outbreak dynamics.
- If the answer is yes, policy that's based on static models will mis-time interventions.

Key Numbers / Facts

! *Two layers of dynamics: (1) weekly rhythm (weekend = 0.6x weekday), (2) behavioral adaptation triggered by prevalence thresholds.*

Possible Questions and Answers

Q: What exactly is wrong with traditional SEIR?

A: Two issues: it assumes homogeneous mixing, and it ignores the behavioral feedback loop. Both lead to overestimating peak severity and underestimating peak timing.

Q: Hasn't dynamic SEIR already been studied?

A: Yes - Funk et al. 2010 and Perra et al. 2011 introduced dynamic-network concepts, but no one has systematically compared static vs dynamic predictions across multiple urban topologies with real-data validation. That gap is what this work closes.

Q: What do you mean by 'urban topology'?

A: The spatial-mixing structure of the city's contact network. I use three regimes: Dense (high mixing, single core), Polycentric (multi-cluster), and Sparse (separated communities).

Slide 3 - Purpose and Objectives

What to Say

- The purpose is to systematically compare static and dynamic epidemic models across three urban spatial regimes and to validate the dynamic model against real COVID-19 data.
- There are five concrete objectives - network generation, dynamic SEIR implementation, running 90 stochastic simulations, real-data validation, and deriving intervention guidelines.
- The five objectives map one-to-one to the six 'task' slides later in the deck.

Key Numbers / Facts

! *90 total runs = 30 stochastic simulations x 3 network types.*

! *Validation against St. Petersburg COVID-19 monitoring data (Winter 2022).*

Possible Questions and Answers

Q: Why 30 simulations per network?

A: 30 is the standard sample size for stochastic SEIR network studies - large enough for stable means and tight confidence intervals, small enough to be tractable on 3 networks.

Q: Why these three city types specifically?

A: They span the spatial-mixing spectrum: from a single dense core to multiple sub-centers to a sparse, segregated layout. Most real cities fall somewhere on that spectrum.

Q: How do you derive 'intervention guidelines' from simulations?

A: By sweeping intervention timing in the model and recording peak reduction, then identifying the day that maximizes reduction for each topology. That gives a per-city optimal window.

Slide 4 - Review of Similar Solutions / Research (1/2)

What to Say

- Network epidemiology has two foundational papers: Pastor-Satorras and Vespignani in 2001 established epidemic thresholds on scale-free networks; Keeling and Eames in 2005 identified degree distribution, clustering, and assortativity as the key drivers of outbreak dynamics.
- On the static side, Rineer et al. produced a synthetic US population of 303 million individuals - impressive scale, but the dataset is static, with no disease dynamics.
- Bilal et al.'s CitySEIRCast integrates a 3D digital twin with agent-based modeling - but it's computationally heavy and limited to a single US county.

Key Numbers / Facts

! *Rineer 2025: synthetic US population via Iterative Proportional Fitting.*

! *Bilal 2025: 3D digital twin + ABM with real mobility data.*

Possible Questions and Answers

Q: Why is Rineer's 303M-person dataset not enough?

A: It's a static snapshot of demographics - no disease simulation, no behavioral adaptation. It tells you who lives where but not how infections would spread or how people respond.

Q: What's Iterative Proportional Fitting?

A: A classical statistical technique to fit a contingency table to known marginals. In synthetic populations, it adjusts a sample so its margins match census counts.

Q: Why not just use CitySEIRCast?

A: It's tied to one specific city's digital twin and computationally too heavy for a comparative study across three topology classes.

Slide 5 - Review of Similar Solutions / Research (2/2)

What to Say

- On the dynamic-network side, Funk et al. introduced behavior-informed networks where infection probability depends on perceived risk. Perra et al. proposed activity-driven networks where node activity changes over time.
- Zhang et al. (2025) provides a conceptual review of ABM vs compartmental/network models but lacks empirical validation.
- The gap that motivates this work: no prior study systematically compares static vs dynamic predictions across multiple urban topologies, with both weekly rhythms AND behavioral adaptation, and validates against real data. All three together - that's what's new.

Key Numbers / Facts

! *Identified Gap: multi-topology + dual dynamics + real validation - none combined before.*

Possible Questions and Answers

Q: Why isn't activity-driven (Perra) already enough?

A: Perra captures temporal activity variation but doesn't model behavioral feedback from awareness of infection levels. Our model adds prevalence-triggered contact reduction on top of an activity rhythm.

Q: Why both weekly patterns AND behavioral adaptation?

A: They operate at different timescales. Weekly rhythm is automatic and recurring (weekday vs weekend); behavioral adaptation is reactive and conditional on perceived risk. Together they capture both regular and crisis-driven contact changes.

Slide 6 - Research Methods and Tools

What to Say

- Method-wise, I generate synthetic contact networks via a Stochastic Block Model with three node types: 435 Social, 75 Hub, 490 Non-social - total 1,000 nodes.
- Spatial mixing is controlled by a Fermi-Dirac distance function with three mu values - 5, 10, 15 - producing Sparse, Polycentric, and Dense regimes.
- On top of these networks I run a dynamic SEIR model with weekly multipliers and threshold-based behavioral adaptation, and I compare against static baselines using paired t-test, ANOVA, and Cohen's d effect size.
- Tools are standard Python scientific stack - NetworkX, NumPy, SciPy, Matplotlib, with Gephi and Plotly for visualization.

Key Numbers / Facts

! *SBM blocks: Social 435 / Hub 75 / Non-social 490.*

! *Fermi-Dirac mu values: 5 (Sparse), 10 (Polycentric), 15 (Dense).*

Possible Questions and Answers

Q: Why a Stochastic Block Model rather than Barabasi-Albert?

A: SBM gives explicit community structure - more realistic for urban populations than pure preferential attachment, which produces hubs but no community modularity.

Q: What is the Fermi-Dirac distance function?

A: A smooth probability function that decays with distance. The mu parameter sets where the transition happens, controlling how 'spread out' contacts are spatially.

Q: Why these specific software libraries?

A: NetworkX is the standard for network science in Python - well-tested and fast at this scale. Gephi and Plotly handle the visualization for publication-quality figures.

Q: How big is the network and why 1,000 nodes?

A: 1,000 nodes per city - chosen for tractability across 90 stochastic runs. The relative comparisons between topologies are robust to N; the absolute attack rates would scale up.

Slide 7 - Scientific Novelty

What to Say

- Four contributions. One: this is the first systematic comparison of static vs dynamic models across three urban spatial regimes.
- Two: a unified framework that integrates Stochastic Block Models, weekly rhythms, and threshold-based behavioral adaptation - typically these have been studied separately.
- Three: real-data validation against St. Petersburg COVID-19 monitoring data with R-squared of 0.915.
- Four: practical, topology-aware intervention guidelines. Public health planning can move from one-size-fits-all to city-specific strategies, approximately doubling effectiveness.

Key Numbers / Facts

! *Tailored strategies are ~2x more effective than one-size-fits-all approaches.*

Possible Questions and Answers

Q: How is this different from Bilal's CitySEIRCast?

A: CitySEIRCast does one city in fine detail with heavy ABM. My work does three topology classes with lightweight SBM plus dynamic SEIR and validates against real data. Different goal: generalization across topologies rather than one-city precision.

Q: What's the practical takeaway for public health?

A: Don't apply uniform lockdowns. In a Dense city you need to act by day 20; in a Sparse city you can wait until day 45. Treating them the same wastes social capital or loses the window.

Q: Why claim '2x effectiveness'?

A: Average peak reduction with the topology-tailored timing is roughly double what you get applying a single universal timing across all three cities.

Slide 8 - The Solution Flow

What to Say

- The workflow has four stages. Stage 1 - build the three SBM-based networks. Stage 2 - run the dynamic SEIR with weekend multipliers and three intervention intensities.
- Stage 3 - run the static baseline on identical networks for direct comparison. Stage 4 - compare static vs dynamic metrics, test significance with ANOVA, and validate against real St. Petersburg data.
- Each stage feeds the next, and the results from each show up in slides 9 through 14.

Key Numbers / Facts

! *Weekend = 60% of weekday contacts. Behavioral reductions: 30% / 50% / 70%.*

Possible Questions and Answers

Q: Why include a static baseline if we already know dynamics matter?

A: Because the question is *how much* they matter quantitatively. Without a controlled static comparison on the same networks, you can't isolate the dynamic effect.

Q: Why three intervention intensities specifically?

A: Mild (30%), Moderate (50%), Strong (70%) span the realistic range of policy responses - from voluntary distancing to soft lockdown to hard lockdown.

Slide 9 - Task 1 - Network Generation Details

What to Say

- Figure 1 shows the three networks at mu equals 15, 10, and 5 - Dense, Polycentric, Sparse.
- All three have 1,000 nodes but very different connectivity. The Dense network has around 49 average degree and low clustering 0.31 - one big mixed core.
- The Sparse network has 16.5 average degree and 0.67 clustering - tight communities, few long-distance contacts. Polycentric sits in the middle.
- These structural differences are what drive the differences in outbreak dynamics later.

Key Numbers / Facts

! *Dense: 49 avg degree, 0.31 clustering*

! *Polycentric: 31 avg degree, 0.42 clustering*

! *Sparse: 16.5 avg degree, 0.67 clustering*

Possible Questions and Answers

Q: Why does sparse have high clustering?

A: Few long-range edges - most contacts stay within tight local communities, which closes many triangles within those communities.

Q: Is this representative of real cities?

A: Dense matches Manhattan-like layouts, Polycentric matches cities like Moscow or LA with multiple sub-centers, Sparse matches suburban or smaller regional cities.

Q: How is 'edges' chosen?

A: Emerges from the SBM block parameters and the Fermi-Dirac spatial function - we tune block connectivity to land on realistic average-degree targets.

Slide 10 - Task 2 - Dynamic SEIR Model

What to Say

- Disease parameters are calibrated to COVID-19 wild-type: R0 of 2.5, incubation 5.2 days, recovery 7 days, transmission rate beta of 0.28 per day. Simulation runs 120 days.
- Two dynamic layers on top. First, a weekly multiplier - weekend contacts are 60% of weekday. Second, behavioral adaptation triggered by prevalence: below 1% nothing changes, at 1-5% contacts drop 30%, at 5-10% they drop 50%, above 10% they drop 70%.
- Each network is simulated 30 times stochastically - 90 runs in total.

Key Numbers / Facts

! *R0 = 2.5, beta = 0.28/day, sigma = 1/5.2/day, gamma = 1/7.0/day, I0 = 50 (0.1%), T = 120 days.*

Possible Questions and Answers

Q: Why R0 = 2.5?

A: Original COVID-19 wild-type estimate from Li et al. 2020 - the New England Journal estimate from Wuhan early transmission data.

Q: Why threshold-based behavioral adaptation rather than continuous?

A: Thresholds capture the way real people respond to news and risk signals - they don't smoothly track prevalence, they react when it crosses noticeable levels. Also keeps the model interpretable.

Q: Are weekend multipliers realistic?

A: Yes - Google COVID-19 Community Mobility Reports show roughly 30-40% reductions on weekends in workplaces and public transit. 0.6 is in that range.

Q: Where do the 30/50/70% reductions come from?

A: Calibrated against published lockdown mobility data. Mild matches voluntary distancing; moderate matches soft lockdowns; strong matches Wuhan-style hard lockdown.

Slide 11 - Task 3 - Static vs Dynamic Comparison

What to Say

- Figure 2 compares static (navy solid) vs dynamic (red dashed) outbreak curves for all three city types. In every panel the dynamic model peaks later and lower than the static one.
- For Dense cities the peak shifts from day 42 to day 54 - an 11.7 day delay - and the attack rate drops from 72% to 58%, a 14% absolute reduction.
- Polycentric and Sparse show smaller but still significant shifts. Statistical tests: paired t-test p equals 0.041, and Cohen's d of 1.96, which is a 'large' effect size.
- Bottom line: static models systematically overestimate peak severity and underestimate peak timing.

Key Numbers / Facts

! *Peak shifts: Dense +11.7d, Polycentric +10.8d, Sparse +7.3d.*

! *Attack-rate reductions: 14.3%, 11.8%, 6.5%.*

! *Cohen's d = 1.96 (large effect).*

Possible Questions and Answers

Q: Why are Sparse cities less affected by dynamics?

A: Slower spread means infection levels rise more slowly, so they stay below the behavioral-trigger thresholds longer. Behavior changes less, so the dynamic effect is smaller.

Q: Is t(2) = 4.82 reliable with only 2 degrees of freedom?

A: Paired t with three cities is admittedly underpowered. We back it up with Cohen's d and ANOVA across the per-run replicates - the conclusion is robust across tests.

Q: Are these results from a single seed or averaged?

A: Means across 30 stochastic runs per network - that's what's plotted and tabled.

Q: Could the static model just be re-calibrated to match?

A: It could be tuned to match a single observed peak, but it cannot reproduce the trajectory shape - the curve flattening and the timing shift are signatures of behavioral feedback.

Slide 12 - Task 4 - Behavioral Scenario Analysis

What to Say

- Figure 3 sweeps intervention intensity on the Dense city. No intervention peaks at day 42 at 18.4%. Mild 30% reduction shifts peak to day 48 at 15.2%. Moderate 50% to day 54 at 12.8%. Strong 70% to day 60 at 10.5%.
- The key insight is diminishing returns. Moderate distancing captures about 72% of the benefit of strong distancing at half the societal cost.
- For Dense cities, moderate is the cost-effective sweet spot.

Key Numbers / Facts

! *Mild->Moderate gains 7 percentage points; Moderate->Strong only gains 5.7.*

! *Attack rate at moderate: 58%, strong: 52% - just 6% difference for 20% more cost.*

Possible Questions and Answers

Q: Why is moderate 'cost-effective'?

A: Strong 70% reduction means major unemployment, supply chain disruption, mental health cost. Moderate captures most of the epidemiological benefit without those second-order costs.

Q: Are these scenarios applied uniformly throughout the simulation?

A: Yes - in this analysis the reduction is the maximum intensity, triggered once thresholds are crossed. Optimal *timing* is analyzed separately in Task 5.

Q: Could combined strategies beat single-intensity?

A: Outside scope, but tiered interventions (mild early, moderate later) would be a logical follow-up.

Slide 13 - Task 5 - Optimal Intervention Timing

What to Say

- Figure 4 shows peak reduction as a function of intervention day, for each city type.
- Dense cities peak at 45% reduction when you act by day 18. Polycentric: 35% at day 28. Sparse: 38% at day 42.
- The lesson is that timing has to be city-specific. Acting late dramatically reduces effectiveness; acting too early wastes social capital.
- Combining city-type with optimal timing gives the tailored strategies on the next slide - roughly twice as effective as a one-size-fits-all policy.

Key Numbers / Facts

! *Optimal days: Dense=18, Polycentric=28, Sparse=42.*

! *Max peak reductions: 45%, 35%, 38%.*

Possible Questions and Answers

Q: Why does Dense need such early action?

A: Faster spread means infections compound quickly. Once past the inflection point, the outbreak has too much momentum to dampen substantially.

Q: How is 'optimal' defined precisely?

A: We simulate intervention activation on each candidate day and measure resulting peak height. Optimal is the day that maximizes peak reduction.

Q: Why is Sparse optimal at day 42 with 38% reduction higher than Polycentric's 35%?

A: Sparse has slower initial spread, so the model has more time to apply the intervention before reaching the inflection - the absolute peak reduction is correspondingly larger even though the curve was already milder.

Slide 14 - Task 6 - Model Validation (St. Petersburg)

What to Say

- To check that this model is realistic, I validated it against St. Petersburg COVID-19 monitoring data from Winter 2022 - the Omicron wave.
- Figure 5 left: normalized outbreak curves. Real data in black, model in blue dashed. Tight alignment. Right: predicted vs observed scatter with R-squared 0.915.
- Auxiliary metrics: RMSE 0.087, MAE 0.065, Pearson r 0.957 - all confirming excellent fit.
- The static baseline only achieves R-squared 0.62 against the same data, which confirms dynamics are required for realism.

Key Numbers / Facts

! *R^2 = 0.915 dynamic vs 0.62 static. Peak timing error < 4 days.*

Possible Questions and Answers

Q: Why St. Petersburg specifically?

A: Open monitoring data, dense urban core relevant to the Dense-city regime, well-documented Omicron wave with clean rise-and-fall shape suitable for curve fitting.

Q: Why isn't R-squared = 1?

A: Real world has factors the model doesn't include: vaccination uptake, variants emerging mid-wave, weather, holidays, testing-capacity bias. 0.915 with a model that uses no fitted parameters for SPB is a strong result.

Q: What's RMSE vs MAE?

A: RMSE penalizes large errors more (squared); MAE is a flat average of absolute errors. Their closeness here means errors are evenly distributed - no big outliers.

Q: Did you fit the model to St. Petersburg data?

A: No. The parameters come from literature, not local fitting. That's why R-squared 0.915 is meaningful - it's an out-of-sample agreement.

Slide 15 - Discussion - Key Findings and Recommended Strategies

What to Say

- Pulling it all together: four key findings.
- One - dynamics matter. Static models overestimate peak severity by 14-20% and underestimate timing by 8-16 days.
- Two - diminishing returns on behavioral intensity. Moderate distancing captures most of the benefit at half the cost.
- Three - timing is city-specific. Dense by day 20, Sparse by day 45.
- Four - validation succeeds. R-squared 0.915 against St. Petersburg confirms the model captures real-world adaptation.
- The bar chart visualizes the magnitude of the dynamic effect across the three cities and the three metrics. Dense is hit hardest by dynamics on every measure.
- The table below summarizes the three recommended strategies that come out of this work.

Key Numbers / Facts

! *Dense: target transport hubs before day 20 -> 40-50% reduction.*

! *Polycentric: synchronized multi-center response before day 28 -> 35-40%.*

! *Sparse: community-based containment before day 42 -> 35-40%.*

Possible Questions and Answers

Q: How would policymakers actually use these strategies?

A: Classify their city's topology (you can do this with mobility data or census-level contact estimates), then look up the recommended strategy and timing window. Strategy is high-level; local public-health authorities specify the operational details.

Q: Have these strategies been tested in real interventions?

A: These are model-derived recommendations, not yet tested in real interventions. Validating them would require a natural experiment or retrospective analysis of cities that implemented different timings.

Q: Why is Dense more sensitive to dynamics than Sparse?

A: Dense networks reach high prevalence quickly, so the behavioral thresholds trigger early and bite hard. Sparse cities never get to high enough prevalence to trigger the strongest behavioral response.

Slide 16 - Conclusion

What to Say

- To conclude: I built and validated a dynamic SEIR framework on three urban topology classes, ran 90 stochastic simulations, and produced topology-aware intervention guidelines.
- Dynamic models predict 8-16 day peak delays and 6-14% attack-rate reductions vs static baselines. Moderate distancing is the cost-effective sweet spot. Optimal timing ranges from day 20 (Dense) to day 45 (Sparse). R-squared 0.915 against St. Petersburg.
- Limitations: the network structure itself is static - only edge weights change. Behavioral parameters come from literature, not local calibration. Weekly patterns are binary weekday/weekend. No age structure.
- Future work: calibrate behavioral thresholds with St. Petersburg mobility data, add age-structured contact matrices, integrate a real-time dashboard, and explore reinforcement learning for adaptive policy discovery.

Key Numbers / Facts

! *90 simulations. R^2 = 0.915. Strategies span 4-week intervention windows by city.*

Possible Questions and Answers

Q: What's the biggest limitation?

A: Network structure is fixed - in reality social networks rewire (people quit jobs, kids change schools, friend groups change). Capturing that would require an adaptive-topology extension.

Q: What's the reinforcement-learning idea?

A: Train an agent to choose intervention strength and timing as a function of observed state (prevalence, hospital load, mobility). It could discover policies better than the fixed threshold rule, and adapt to new variants or new cities.

Q: Why no age structure?

A: Adding age would require an age-mixing matrix and per-age severity/recovery parameters - doable but doubles the parameter space. It's on the short-term roadmap.

Q: Is this ready for publication?

A: Medium-term plan is journal submission once age structure and a second-city validation are in. The core framework is publication-ready.

Slide 17 - Thank You

What to Say

- Thank you for your attention. I welcome questions on any part of the methodology, results, or implications.

Possible Questions and Answers

Q: What's the simplest one-sentence summary?

A: Static epidemic models systematically misjudge urban outbreaks - adding weekly rhythms and behavioral adaptation gives topology-specific intervention windows that approximately double effectiveness.

Q: How can people follow up?

A: Through ITMO University - contact via my university email, or through Prof. Leonenko's research group.